

UNIVERSIDAD NACIONAL DE SAN AGUSTÍN

FACULTAD DE INGENIERÍA DE PRODUCCIÓN Y SERVICIOS

Escuela Profesional de Ingeniería Electrónica



INTRODUCCIÓN AL DESARROLLO DE NUEVAS PLATAFORMAS

“Informe 03”

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DOCENTE:

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Sección: “A”

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Código: MainActivity

Kotlin

```
package com.example.compras

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.activity.enableEdgeToEdge
import androidx.compose.foundation.layout.fillMaxSize
import androidx.compose.foundation.layout.padding
import androidx.compose.material3.Scaffold
import androidx.compose.material3.Text
import androidx.compose.runtime.Composable
import androidx.compose.ui.Modifier
import androidx.compose.ui.tooling.preview.Preview
import com.example.compras.ui.theme.ComprasTheme
import com.example.compras.ui.screens.Principal
import androidx.compose.material3.MaterialTheme
import androidx.compose.material3.Surface

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContent {
            ComprasTheme {
                Surface(
                    modifier = Modifier.fillMaxSize(),
                    color = MaterialTheme.colorScheme.background
                ) {
                    // Llamamos a la pantalla principal que creamos en el paso
                    anterior
                    Principal()
                }
            }
        }
    }
}

@Composable
fun Greeting(name: String, modifier: Modifier = Modifier) {
    Text(
        text = "Hello $name!",
        modifier = modifier
    )
}
```

```

}

@Preview(showBackground = true)
@Composable
fun GreetingPreview() {
    ComprasTheme {
        Greeting("Android")
    }
}

```

Screen Principal

```

Kotlin
package com.example.compras.ui.screens

import androidx.compose.foundation.layout.Arrangement
import androidx.compose.foundation.layout.Column
import androidx.compose.foundation.layout.Row
import androidx.compose.foundation.layout.Spacer
import androidx.compose.foundation.layout.fillMaxSize
import androidx.compose.foundation.layout.height
import androidx.compose.foundation.layout.padding
import androidx.compose.material3.Button
import androidx.compose.material3.Checkbox
import androidx.compose.material3.MaterialTheme
import androidx.compose.material3.Scaffold
import androidx.compose.material3.Text
import androidx.compose.runtime.Composable
import androidx.compose.runtime.getValue
import androidx.compose.runtime.mutableStateOf
import androidx.compose.runtime.remember
import androidx.compose.runtime.setValue
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.unit.dp

@Composable
fun Principal(modifier: Modifier = Modifier) {
    // Elevación de estado (State Hoisting) para compartir entre composables
    var count by remember { mutableStateOf(1) }
    var isCanchita by remember { mutableStateOf(false) }
    var isBebida by remember { mutableStateOf(false) }
    var isCupon by remember { mutableStateOf(false) }

```

```

val precioEntrada = 10
val precioCanchita = 5
val precioBebida = 3
val precioCupon = 2
Scaffold(modifier = modifier) { paddingValues ->
    Column(
        modifier = Modifier
            .fillMaxSize()
            .padding(paddingValues)
            .padding(16.dp),
        horizontalAlignment = Alignment.CenterHorizontally,
        verticalArrangement = Arrangement.Center
    ) {
        Head()

        Spacer(modifier = Modifier.height(24.dp))

        Entradas(
            count = count,
            onIncrement = { count++ },
            onDecrement = { if (count > 0) count-- }
        )

        Spacer(modifier = Modifier.height(16.dp))

        Extras(
            isCanchita = isCanchita,
            onCanchitaChange = { isCanchita = it },
            isBebida = isBebida,
            onBebidaChange = { isBebida = it }
        )
        Cupon(
            isCupon = isCupon,
            onCuponChange = { isCupon = it }
        )

        Spacer(modifier = Modifier.height(24.dp))

        Total(
            count = count,
            precioEntrada = precioEntrada,
            isCanchita = isCanchita,
            precioCanchita = precioCanchita,
            isBebida = isBebida,
            precioBebida = precioBebida,
            isCupon = isCupon,
            precioCupon = precioCupon
        )
    }
}

```

```

        )
    }
}

```

```

@Composable
fun Head(modifier: Modifier = Modifier) {
    Text(
        text = "Compra de Entradas",
        style = MaterialTheme.typography.headlineMedium,
        modifier = modifier
    )
}

```

```

@Composable
fun Entradas(
    count: Int,
    onIncrement: () -> Unit,
    onDecrement: () -> Unit,
    modifier: Modifier = Modifier
) {
    Column(horizontalAlignment = Alignment.CenterHorizontally, modifier =
modifier) {
        Text(text = "Cantidad de entradas")
        Spacer(modifier = Modifier.height(8.dp))
        Row(
            verticalAlignment = Alignment.CenterVertically,
            horizontalArrangement = Arrangement.spacedBy(16.dp)
        ) {
            Button(onClick = onDecrement) {
                Text("-")
            }
            Text(
                text = "$count",
                style = MaterialTheme.typography.titleMedium
            )
            Button(onClick = onIncrement) {
                Text("+")
            }
        }
    }
}

```

```

@Composable
fun Check(
    checked: Boolean,
    onCheckedChange: (Boolean) -> Unit,
    label: String,

```

```

        modifier: Modifier = Modifier
    ) {
        Row(
            verticalAlignment = Alignment.CenterVertically,
            modifier = modifier
        ) {
            Checkbox(
                checked = checked,
                onCheckedChange = onCheckedChange
            )
            Text(text = label)
        }
    }
}

@Composable
fun Extras(
    isCanchita: Boolean,
    onCanchitaChange: (Boolean) -> Unit,
    isBebida: Boolean,
    onBebidaChange: (Boolean) -> Unit,
    modifier: Modifier = Modifier
) {
    Column(horizontalAlignment = Alignment.Start, modifier = modifier) {
        Text(
            text = "Adicionales:",
            style = MaterialTheme.typography.titleMedium
        )
        Check(
            checked = isCanchita,
            onCheckedChange = onCanchitaChange,
            label = "Canchita ($5)"
        )
        Check(
            checked = isBebida,
            onCheckedChange = onBebidaChange,
            label = "Bebida ($3)"
        )
    }
}

@Composable
fun Cupon (
    isCupon: Boolean,
    onCuponChange: (Boolean) -> Unit,
    modifier: Modifier = Modifier
){
    Check(
        checked = isCupon,

```

```

        onCheckedChange = onCuponChange,
        label = "Cupon de descuento ($2)"
    )
}

@Composable
fun Total(
    count: Int,
    precioEntrada: Int,
    isCanchita: Boolean,
    precioCanchita: Int,
    isBebida: Boolean,
    precioBebida: Int,
    isCupon: Boolean,
    precioCupon: Int,
    modifier: Modifier = Modifier
) {
    val subtotalEntradas = count * precioEntrada
    val subtotalCanchita = if (isCanchita) precioCanchita else 0
    val subtotalBebida = if (isBebida) precioBebida else 0
    val descuento = if (isCupon) precioCupon else 0
    val montoSubTotal = subtotalEntradas + subtotalCanchita + subtotalBebida
    val montoTotal = montoSubTotal - descuento

    Text(
        text = "Total a pagar: $$montoSubTotal",
        style = MaterialTheme.typography.titleLarge,
        modifier = modifier
    )
    Text(
        text = "descuento: $$descuento",
        style = MaterialTheme.typography.titleLarge,
        modifier = modifier
    )
    Text(
        text = "Total a pagar: $$montoTotal",
        style = MaterialTheme.typography.titleLarge,
        modifier = modifier
    )
}

```

9:13



Compra de Entradas

Cantidad de entradas



Adicionales:

☒ Canchita (\$5)

☐ Bebida (\$3)

☒ Cupon de descuento (\$2)

Total a pagar: \$35

descuento: \$2

Total a pagar: \$33



Renderizado de interfaz